EE345 Mod4-Lec1: Classification via Logistic Regression

[VMLS, 14]

Classification

- **So far...**: goal was to predict a real-valued outcome y from some data x
- **Module 4** (Classification): the outcome y takes on only a finite number of values, and hence is sometimes called a **label**, or in statistics, a **categorical**.

Binary Classification

- As before, the goal is to predict the outcome y from the feature vector x.
- The classifier f is a function that takes in a feature vector x and outputs a prediction $f(x)$.
- But now the outcome y takes on only two values:

$$\text{typically } \{-1, +1\} \quad \text{or} \quad \{0, 1\}$$

From Regression to Classification

- So far: **least squares** for regression problems

$$y \approx \theta^\top x, \quad y \in \mathbb{R}.$$

- Now: **binary classification**

- Idea: keep a linear score, then pass it through a *squashing* function to get a probability.

Classification Examples

- **Email spam detection**.
  - *Feature vector*: $x \in \mathbb{R}^n$ contains features of an email message like word counts etc.
  - *Outcome*: $y = +1$ if an email represented by feature vector x is `SPAM` and -1 otherwise.
- **Fraud detection.**
  - *Feature vector*: $x \in \mathbb{R}^n$ contains features associated with a credit card user such as average monthly spending, median prices of purchases over last week, etc.
  - *Outcome*: $y = +1$ for `fradulent transactions`, and -1 otherwise.
- **Document Classification.**
  - *Feature vector*: $x \in \mathbb{R}^n$ is a word count (or histogram) vector for a document
  - *Outcome*: $y = +1$ if the document has some topic (e.g., politics) and -1 otherwise

Two Linear Approaches to Binary Classification

- We consider $y \in \{-1, +1\}$ and a linear score:

$$s(x) = \theta^\top x.$$

- Two ways to turn scores into predictions:
  - **Least Squares Classification**: choose θ to make $s(x_i)$ close to y_i.
  - **Logistic Regression**: choose θ to model $p(y_i = 1 \mid x_i)$ via logistic function.

Method 1: Least Squares Classification

Least Squares Classifier

- Least squares classifier: this is a simple method that works well in many cases
- do ordinary real-valued least squares fitting of the outcome, ignoring that $y \in \{-1, +1\}$
- i.e., choose basis functions $\phi_1, \ldots, \phi_p$ and solve

- **final classifier (least squares classifier)**:

Intuition for Least Squares Classifier

- The value $\tilde{f}(x)$ is a number "near" $+1$ when $y = +1$ and near -1 when $y = -1$
- Forced to guess one of the two possible outcomes, $\text{sign}(\tilde{f}(x))$ is a good choice—it is the nearest neighbor of $\tilde{f}(x)$ among $\{-1, +1\}$
- $\tilde{f}(x)$ also tells us our confidence in our assignment

Least Squares Classification

- If we pick $\phi_1(x) = 1$ and $\phi_2(x) = x$, then the problem is

$$\min_{\theta} \|X\theta - y\|_2^2, \quad \text{where } y_i \in \{-1, +1\}.$$

- Closed-form solution:

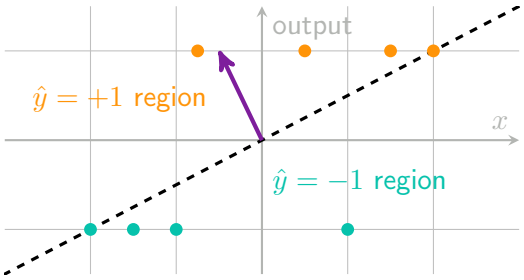
$$\hat{\theta}_{\mathsf{LS}} = (X^\top X)^{-1} X^\top y.$$

- Prediction rule:

- Pros:
  - Simple, fast, closed-form.
  - Works surprisingly well when classes are linearly separable.

- Cons:
  - Squared error is not made for classification.
  - Large errors are overly penalized (outliers).
  - Predicted scores are not probabilities.

Least Squares Classifier with the Sign Function



- Map labels to $\{-1, +1\}$ and fit:

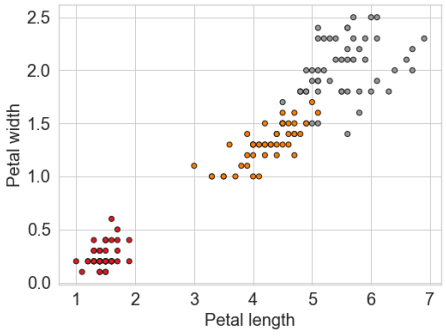
$$\hat{\theta} = \arg\min_{\theta} \sum_{i} (y_i - \theta_0 - \theta_1 x_i)^2.$$

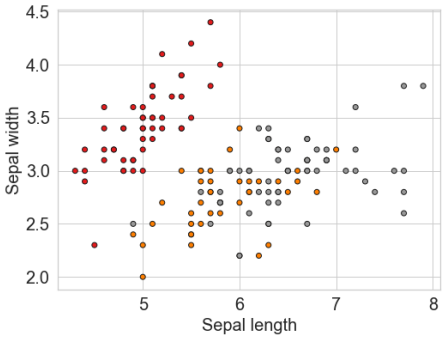
- Prediction is a **real-valued** output

- Turn it into a classifier with

Example: Predicting Iris Type

Three types of iris—`setosa`, `versicolour`, `virginica`—50 samples of each type





four features:

- x_1 `sepal length [cm]`, x_2 `sepal width [cm]`
- x_3 `petal length [cm]`, x_4 `petal width [cm]`

- Goal: build classifier to detect if iris is `virginica` or not

Iris Data Set Example: Confusion Matrix

Method 2: Logistic Regression

Logistic Regression: Fixing the Problems

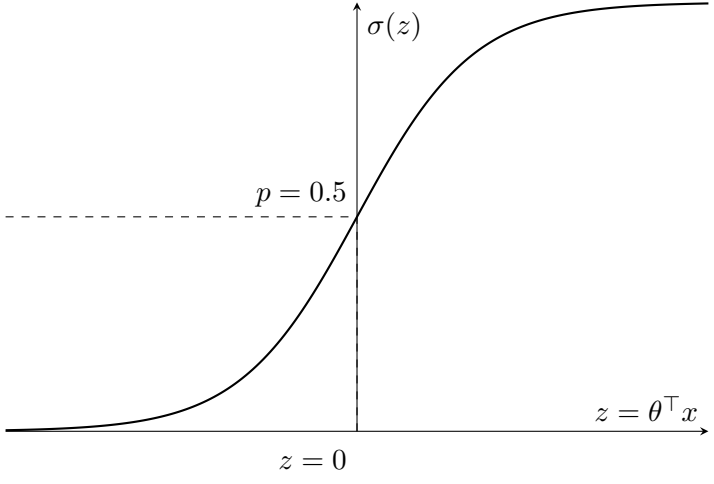
Decision Boundary

Because the sigmoid is monotone:

$$p(y = 1 \mid x) = 0.5 \quad \Longleftrightarrow \quad \theta^\top x = 0.$$

Same boundary as linear regression classification, BUT better calibrated probabilities.

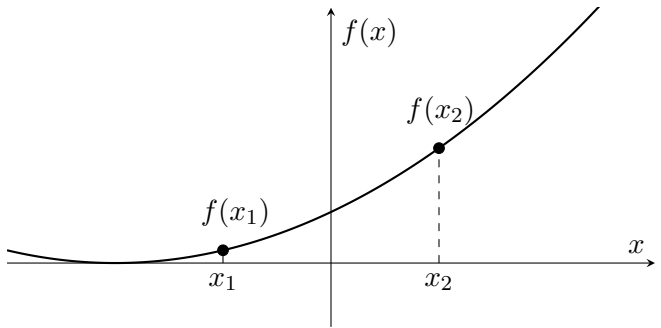
Sigmoid



The linear score $z = \theta^\top x$ is mapped through the sigmoid to a probability $\sigma(z) \in (0, 1)$; $z = 0$ corresponds to $p = 0.5$.

Sigmoid Properties: Monotone functions are order-preserving

A function $f : \mathbb{R} \to \mathbb{R}$ is **monotone increasing** if



Checking Monotonicity Using the Derivative

Monotone increasing: if $f \in C^1$ (i.e. continuous and differentiable)

Strictly monotone increasing: $f'(x) > 0$ for all x

Monotone decreasing: $f'(x) \leq 0$ for all x

Derivative Test: Necessary and Sufficient (for C^1)

Assume: $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable (i.e., $f \in C^1$).
Sufficient:

$$f'(x) \geq 0 \ \ \forall x \quad \Longrightarrow \quad f \text{ is monotone non-decreasing.}$$

Necessary: If f is monotone non-decreasing, then for every x,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \geq 0.$$

Conclusion: For C^1 (and therefore C^2) functions,

$$f \text{ monotone increasing} \quad \Longleftrightarrow \quad f'(x) \geq 0 \ \forall x.$$

Note: Without differentiability, monotonicity does *not* require f' to exist.

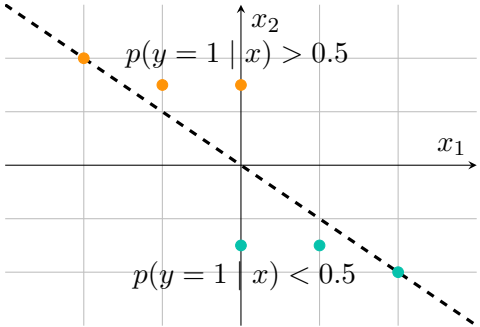
Monotonicity of the Sigmoid

The logistic (sigmoid) function is

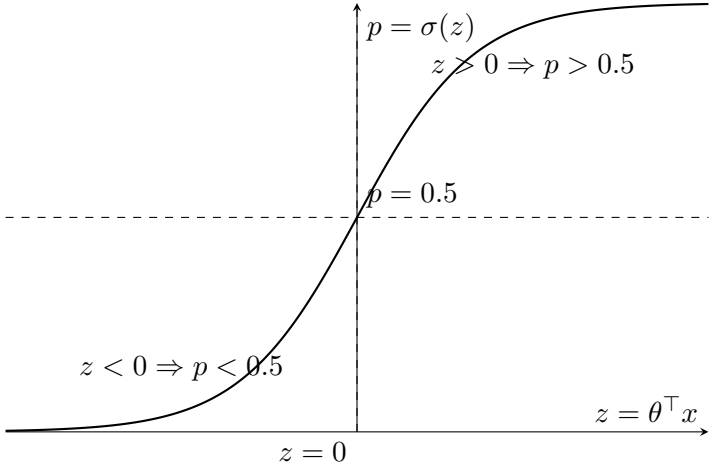
$$\sigma(z) = \frac{1}{1+e^{-z}} \qquad \sigma'(z) = \sigma(z)\big(1-\sigma(z)\big).$$

Decision Boundary for Logistic Regression

$$p(y = 1 \mid x) = \sigma(\theta^\top x).$$



Sigmoid: Mapping $\theta^\top x$ to Probability



Because σ is strictly increasing and $\sigma(0) = 0.5$,

$$\theta^\top x > 0 \implies \sigma(\theta^\top x) > 0.5, \quad \theta^\top x < 0 \implies \sigma(\theta^\top x) < 0.5.$$

Why Can Sigmoid Outputs Be Interpreted as Probabilities?

Odds and Log-Odds

Odds

For a probability $p \in (0, 1)$, the *odds* of the event are

Log-odds (logit)

In logistic regression,

Interpreting Coefficients via Odds Ratios

Consider a single feature x and model

Why Can Sigmoid Outputs Be Interpreted as Probabilities?

- Logistic regression assumes a probabilistic model:

$$y_i \sim \text{Bernoulli}(p_i), \qquad p_i = P(y_i = 1 \mid x_i, \theta).$$

- We model the **log-odds** as linear:

$$\log \frac{p_i}{1 - p_i} = \theta^\top x_i.$$

- Solving for p_i gives the sigmoid:

$$p_i = \sigma(\theta^\top x_i) = \frac{1}{1 + e^{-\theta^\top x_i}}.$$

- Therefore $\sigma(\theta^\top x)$ is the **model's estimate** of the probability that $y = 1$.
- **Calibration:** These probabilities may or may not be well-calibrated depending on the dataset and training.

Fitting the Logistic Regression Model

Summary so far: Logistic Regression

- Model a probability:

$$p(y = 1 \mid x) = \sigma(\theta^\top x) = \frac{1}{1 + e^{-\theta^\top x}}.$$

- Prediction rule:

$$\hat{y}(x) = \begin{cases} 1, & p(x) \geq 0.5, \\ 0, & p(x) < 0.5. \end{cases}$$

- Outputs calibrated probabilities.
- **How do we fit the model?**

Intuition: Cost function via Likelihood

Likelihood: A score of how well our model explains the data we observed.

- High likelihood = the model "expected" the data.
- Low likelihood = the model is "surprised."

For classification: If the true label is $y \in \{0, 1\}$ and the model predicts p, we want:

$$y = 1 \implies p \text{ large}, \qquad y = 0 \implies (1 - p) \text{ large}.$$

Why log likelihood? Turning products into sums makes optimization easier and more stable. Negative log likelihood becomes a loss function.

Cross entropy loss:

$$\ell = -\big(y \log p + (1 - y) \log(1 - p)\big)$$

Cross-entropy punishes confident wrong predictions very strongly, and rewards confident correct predictions.

Likelihood and Loss for Logistic Regression

- Data: (x_i, y_i) for $i = 1, \ldots, n$, with $y_i \in \{0, 1\}$.
- Model:
$$p_i := p(y_i = 1 \mid x_i) = \sigma(\theta^\top x_i).$$
- Likelihood (under independence):
- Log-likelihood:
- We **maximize** $\ell(\theta)$, or equivalently **minimize** the negative log-likelihood:

Loss Function: Negative Log-Likelihood or Cross-Entropy Loss

Maximum likelihood $\implies$ minimize logistic loss:

$$\ell(\theta) = -\sum_{i=1}^{n} \Big[y_i \log(\sigma(\theta^\top x_i)) + (1 - y_i) \log(1 - \sigma(\theta^\top x_i)) \Big] .$$

Properties:

- Convex in θ.
- No closed form $\longrightarrow$ use gradient descent.

Gradient of the Logistic Loss

- Recall $p_i = \sigma(\theta^\top x_i)$ and

$$J(\theta) = -\sum_{i=1}^{n} \Big(y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big).$$

- Using $\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$, one can show

$$\nabla_\theta J(\theta) = \sum_{i=1}^{n} (p_i - y_i)\, x_i.$$

- In matrix notation: let $X \in \mathbb{R}^{n \times d}$ have rows $x_i^\top$, $p \in \mathbb{R}^n$ have entries p_i, and $y \in \mathbb{R}^n$ have entries y_i. Then

$$\nabla_\theta J(\theta) = X^\top (p - y).$$

Gradient of the Logistic Loss: Detailed Derivation (1)

Gradient of the Logistic Loss: Detailed Derivation (2)

Gradient of the Logistic Loss: Detailed Derivation (3)

Gradient of the Logistic Loss: Detailed Derivation (4)

Gradient Descent for Logistic Regression

Logistic Regression with Labels $\{-1, +1\}$

With labels $y_i \in \{-1, +1\}$, the logistic model can be written in a clean, symmetric form.

Model:

$$p(y_i \mid x_i; \theta) = \sigma(y_i\, \theta^\top x_i), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Interpretation:

- If $y_i = +1$, we want $\theta^\top x_i$ to be large and positive
- If $y_i = -1$, we want $\theta^\top x_i$ to be large and negative
- Both cases are captured by $y_i \theta^\top x_i$

Negative log-likelihood:

$$\ell(\theta) = \sum_{i=1}^{n} \log\Big(1 + \exp\left(-y_i \theta^\top x_i\right)\Big).$$

This form is simpler than the $\{0, 1\}$ expression, which requires two separate terms: $y \log p + (1 - y) \log(1 - p)$.

Simpler Gradients with Labels $\{-1, +1\}$

For logistic regression with loss

$$\ell(\theta) = \sum_{i=1}^{n} \log\Big(1 + \exp\left(-y_i \theta^\top x_i\right)\Big),$$

the gradient has a simple unified form.
Derivative with respect to θ:

$$\nabla_\theta \ell(\theta) = \sum_{i=1}^{n} \left(\frac{1}{1 + \exp\left(y_i \theta^\top x_i\right)}\right)(-y_i x_i).$$

Simpler Gradients with Labels $\{-1, +1\}$

Using the sigmoid identity

$$\sigma(-y_i \theta^\top x_i) = \frac{1}{1 + \exp\left(y_i \theta^\top x_i\right)},$$

the gradient becomes

$$\boxed{\nabla_\theta \ell(\theta) = \sum_{i=1}^{n} - y_i \, x_i \, \sigma(-y_i \theta^\top x_i)}$$

Key advantages:

- One unified formula (no separate $y = 1$ / $y = 0$ cases)
- Symmetric treatment of the two classes
- Clean geometry: pushing $y_i \theta^\top x_i$ to be large and positive

Regularized Logistic Regression

- To control overfitting, we can add an ℓ_2 penalty:

$$J_\lambda(\theta) = -\sum_{i=1}^{n} \Big(y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big) + \lambda \|\theta\|_2^2.$$

- Gradient becomes

$$\nabla_\theta J_\lambda(\theta) = X^\top (p - y) + 2\lambda\theta.$$

- Gradient descent update:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \Big(X^\top (p^{(t)} - y) + 2\lambda\theta^{(t)} \Big).$$

- Same idea as ridge regression, but with logistic loss.

Next Time: Prediction Errors: Model Performance Metrics